

# Scalable Inversion of Contests with Correlated Performances, Including Softmax and Multinomial Probit

Peter Cotton\*

August 2026

## Abstract

We give a scalable engine for inference and counterfactual selection in large correlated races, and specialize it to multinomial probit with supplied low-rank-plus-diagonal covariance. Multinomial probit choice probabilities over  $n$  alternatives are Gaussian orthant integrals, and for thirty years econometrics has computed them by simulation, most commonly with the GHK sampler, whose  $R$  draws cost  $O(n^2R)$  per probability,  $n$ -fold that for a full vector, with error  $O(R^{-1/2})$ . We observe that the choice event is one-dimensional: the probability that alternative  $i$  wins is an integral over the winning value, and one shared survival field prices all  $n$  alternatives at once. For covariances under which performances are conditionally independent given a few shared variables (factor, block, nested and hierarchical structures, the restrictions applied multinomial probit imposes in practice), one shared field computes all  $n$  probabilities, matrix-free Jacobian-vector products, and every single-removal counterfactual, in  $O(nLQ)$  operations over a lattice of  $L$  winning values and  $Q$  factor quadrature nodes, and inverts observed probabilities back to utilities under a fixed, known covariance (the Jacobian is exact for factor, block and nested structures, and for trees exact within clusters). That Jacobian is a weighted graph Laplacian whose edge weights are photo-finish tie densities: the geometry of substitution between alternatives, delivered by the same pass that prices them, so removal counterfactuals reuse the field — on a lattice certified to cover the post-removal winner, whose bulk is the first- or second-finisher distribution of the original race (the implementation checks the post-removal mass rather than renormalizing a defect away). At  $n = 200$  it is two hundred times faster than GHK at  $10^4$  draws, with error below the comparison’s resolution; GHK’s measured cost grows superquadratically (exponent 2.15 to 2.8 across the measured range) against the lattice’s near-linear growth, so the ratio widens by orders of magnitude. The same two calls price and invert softmax, Thurstone–Mosteller, factor multinomial probit, and a hierarchy of team and sector models; forward pricing is benchmarked to ten million alternatives. These objects support not only discrete-choice estimation but also selection and routing among large correlated populations of models, experts, or autonomous agents.

## 1 Introduction

A decision maker faces  $n$  alternatives and chooses the one with the highest utility. Write  $U_i = \mu_i + \varepsilon_i$  with  $\varepsilon \sim N(0, \Sigma)$ ; the probability that alternative  $i$  is chosen is

$$p_i = \Pr\{U_i \geq U_j \text{ for all } j\} = \Pr\{\tilde{U} \geq 0\}, \quad \tilde{U}_j = U_i - U_j, \quad (1)$$

---

\*[peter.cotton@microprediction.com](mailto:peter.cotton@microprediction.com). The reference implementation, its parity-locked ports (Python, R, Rust, JavaScript) and the seeded scripts behind every number in this paper live at [github.com/microprediction/winning](https://github.com/microprediction/winning); this revision’s tables were produced at repository tag `paper-r1`.

an  $(n - 1)$ -dimensional Gaussian orthant integral. This is the multinomial probit model. Its appeal has been understood since Thurstone (1927): unlike the logit family it imposes no independence of irrelevant alternatives, because  $\Sigma$  lets alternatives be close substitutes. Its curse has been understood just as long. The integral (1) has no closed form, and estimation needs it, and its derivatives, at every trial value of the parameters.

The workhorse answer is the GHK simulator of Geweke (1989), Börsch-Supan and Hajivassiliou (1993), Keane (1994) and Hajivassiliou et al. (1996). GHK factors  $\Sigma$  by Cholesky and writes the orthant probability as a nested sequence of one-dimensional truncated normal probabilities, each conditioning on draws for the previous coordinates. Averaging  $R$  such importance weights gives an unbiased, smooth estimate of one  $p_i$  at cost  $O(n^2)$  per draw. The estimator made maximum simulated likelihood practical and it remains the default in textbooks (Train, 2009). Its costs are also standard: error shrinks as  $O(R^{-1/2})$ ; the log of a simulated probability is biased at  $O(1/R)$ ; gradients inherit the simulation noise; and a full probability vector costs  $n$  separate runs. A single likelihood contribution needs only the chosen alternative's probability, so the  $n$ -fold cost falls not on the individual log-likelihood but on everything else estimation touches: share inversion, aggregate-share and market models, prediction and welfare across the menu, and the derivative of any of these with respect to all  $n$  utilities.

This paper takes a different route to the same object, in a min-wins convention fixed here once: performances are times,  $X_i = -U_i$ , and the lowest time wins. The choice event is one-dimensional. Contestant  $i$  wins when its time beats the field, so

$$p_i = \int f_i(x) \prod_{j \neq i} S_j(x) dx, \quad (2)$$

where  $f_i$  is the density of  $X_i$  and  $S_j$  the survival of  $X_j$ : integrate over the winning time  $x$ , requiring every rival to come in slower. For independent performances Cotton (2021) computed (2) for all  $n$  on a lattice by building the field  $\sum_j \log S_j$  once and dividing each alternative out, and inverted the map from probabilities back to  $\mu$ . The present paper extends the field construction to correlated performances by conditioning: whenever  $\Sigma$  makes performances independent given a few shared variables, the field factorizes, and all  $n$  probabilities, Jacobian-vector products, and each inversion iteration cost  $O(nLQ)$  for a lattice of  $L$  points and  $Q$  quadrature nodes; the Jacobian itself is a weighted graph Laplacian applied as an operator, never materialized. No per-alternative repetition; error controlled by quadrature, deterministic for the Gauss-Hermite rules and reproducibly randomized for scrambled-Sobol nodes.

Only utility contrasts and a normalized scale are identified in multinomial probit, so the raw covariance is overparameterized, and the contrast covariance may in principle carry unrestricted parameters. Applied work nevertheless imposes structured covariance as a matter of course, for parsimony and for computation (Train, 2009); the grammars here cover the structured families in common use. The table lists them: the first six rows are nested exactly and each is priced and inverted by the same two calls; the last row is a fitted approximation developed in §6.

| model                     | covariance grammar                              | source         |
|---------------------------|-------------------------------------------------|----------------|
| Luce / softmax            | independent, minimum-Gumbel base, $D = \pi^2/6$ | exact identity |
| Thurstone–Mosteller       | independent, normal base                        | Cotton (2021)  |
| factor multinomial probit | $VV' + \text{diag}(D)$                          | §4.1           |
| teams, conferences        | blocks: rank- $r$ effect per cluster            | §4.2           |
| market plus sectors       | nested: blocks $\times$ global factor           | §4.2           |
| hierarchies               | tree: uniform shared effects                    | §4.3           |
| dense $\Sigma$            | fitted grammar + residual factors (approx.)     | §6             |

Scale motivates second. The next table reports measured wall clock for all  $n$  probabilities (Rust kernels, one laptop) against extrapolations of GHK’s measured cost. The measurements behind the law: the complete-vector GHK timings at  $R = 10^3$  are 0.57, 5.66 and 51.3 seconds at  $n = 200, 500$  and 1000 (times at  $R = 10^4$  scale by ten), a log-log slope of 2.8 over that range, while the small- $n$  table below gives 2.15 from  $n = 10$  to 200: the law is superquadratic and steepening. Extrapolations are shown as brackets over the two measured exponents, and every timing is regenerated by the committed `bench.py`. At these budgets GHK’s total variation error is near  $10^{-2}$ ; the lattice error is near  $10^{-8}$ . The bracketed rows extrapolate far beyond anything GHK is actually run at; that is their point. Nobody prices a million-alternative field with GHK, and the brackets say what the cost law implies about why.

| $n$    | lattice, measured | GHK at $10^4$ draws, cost-law bracket | multiplier                      |
|--------|-------------------|---------------------------------------|---------------------------------|
| $10^4$ | 0.18 s            | 20 hours – 3.7 days                   | $4 \times 10^5 - 2 \times 10^6$ |
| $10^5$ | 2.7 s             | 0.3 – 6.5 years                       | $4 \times 10^6 - 8 \times 10^7$ |
| $10^6$ | 29 s              | 46 – 4,100 years                      | $5 \times 10^7 - 4 \times 10^9$ |

Section 2 places the method among the alternatives. Section 3 gives the independent lattice race; Section 4 extends it across the covariance grammar, factor to blocks to trees, each with its recursion written out. Section 5 gives the Jacobian operator and the inversion. Section 6 reduces dense covariance to the grammar family, approximately, and reports accuracy per named random-correlation ensemble. Section 7 contains the benchmarks.

## 2 The alternatives

*Deterministic quadrature.* Genz (1992) transformed the orthant integral to the unit cube and integrated by adaptive and quasi-Monte Carlo rules; the `mvtnorm` implementation is the standard in R. Accuracy is not the issue: in the benchmark below it agrees with the lattice to the reference’s own noise. The shape is the issue. It treats one probability at a time, so the all- $n$  vector at  $n = 30$  costs three hundred times the lattice, and the gap widens with  $n$ .

*Frequency simulation.* The crude frequency estimator of Lerman and Manski (1981) counts argmax draws. It is unbiased for the whole vector at once but not smooth in parameters, and its per-entry relative error explodes for small  $p_i$ . Measured below: at wall clock matched to the lattice it carries twelve to twenty times the total-variation error, records zero counts on tail alternatives at  $n = 30$ , and at ten times the budget remains a factor of two to four behind.

*GHK.* Defined above. Its dominance is documentable rather than folklore. Stata’s `cmmprobit` manual states that the likelihood evaluator “implements the Geweke–Hajivassiliou–Keane (GHK) algorithm to approximate the multivariate distribution function” (StataCorp, 2023; Gates, 2006), and the command’s entire `intmethod` menu (Hammersley, Halton, or pseudo-random point sets) selects only the sequence feeding the same simulator: the market-leading implementation offers no non-GHK method at all. R’s `mlogit` and `mvProbit` default to GHK likewise (Croissant, 2020), and the GHK-based `mvprobit` of Cappellari and Jenkins (2003) carried the method to applied Stata work at large. Two properties matter for comparison: it is smooth in parameters, which frequency counting is not, and it prices one alternative per run. Stern (1992) gave a related decomposition with an analytic leading term.

*Bayesian data augmentation.* Albert and Chib (1993) and McCulloch and Rossi (1994) sidestep the integral entirely by Gibbs sampling the latent utilities; Imai and van Dyk (2005) implement the marginal data-augmentation version in the MNP package, and Loaiza-Maya and Nibbering (2022) carry the Bayesian route to large choice sets by variational methods. For Bayesian inference this is a complete and often excellent answer. It is simply answering a different question than the one benchmarked here: the cost moves into the Markov chain, the likelihood surface is never formed, and choice probabilities appear only as posterior functionals, so there is no pricing call to time or to check.

*Analytic approximations.* Clark (1961) matched moments of pairwise maxima; Mendell and Elston (1974) and Solow (1990) approximate by sequential conditioning on first and second moments. These are fast, sometimes startlingly accurate, and carry no error control, and the benchmark below measures all three properties at once: Mendell–Elston prices the  $n = 10$  vector in two milliseconds within  $2 \times 10^{-3}$  of the reference, then drifts to  $3 \times 10^{-2}$  at  $n = 30$  with tail probabilities off by a factor of nearly four, and nothing in the output announces the change. The Clark-type recursion behaves alike. The family is current practice, not history: Bhat (2011) builds the MACML estimator on precisely these approximations, and the `Rprobit` package implements it (Bauer, 2024), with its asymptotics examined by Batram and Bauer (2019). Expectation propagation is the modern member of the same family. We benchmark representative sequential moment approximations through the classical members; MACML’s composite-likelihood construction around its analytic ingredient is its own estimator and is not itself benchmarked here.

*Minimax tilting.* Botev (2017) computes single Gaussian rectangle probabilities with bounded relative error deep into the tails, in dimensions up to roughly a thousand. It is the accuracy reference, and we use it as the referee for our own tail claims; it prices one probability per run.

Every smooth, likelihood-grade method above prices one probability at a time; frequency counting obtains the vector but not derivatives. A single likelihood contribution is content with one probability; the tasks that surround it are not. Share inversion, prediction across the menu, welfare, and derivatives with respect to all  $n$  utilities each want the whole vector at every parameter value. Sharing the field across alternatives is the algorithmic point of this paper, and it is what makes the  $n$ -fold and the derivative costs disappear for those tasks.

### 3 The lattice race

In the min-wins convention of the introduction, let  $X_i = \mu_i + \sigma_i \epsilon_i$  with  $\epsilon_i$  i.i.d. from a standardized base density (normal, Gumbel, or any density supplied as code), and write  $f_i, S_i$  for the density and survival of  $X_i$ . One notational warning, once and loudly: from here on  $\mu_i$  is a TIME location — raising it makes contestant  $i$  worse and lowers  $p_i$  — the negative of the introduction’s utility location. Inversion returns time locations; utility coefficients are their negation, and implementations that forget the sign flip invert their leaderboards.

The algorithm evaluates (2) on a lattice  $x_1 < \dots < x_L$ :

1. compute  $\log S_j(x_\ell)$  and  $\log f_j(x_\ell)$  for all  $j, \ell$ ;
2. accumulate the field  $L(x_\ell) = \sum_j \log S_j(x_\ell)$ ;
3. for each  $i$ , form the integrand  $f_i(x_\ell) e^{L(x_\ell) - \log S_i(x_\ell)}$ , which is contestant  $i$ ’s density times everyone else’s survival, with  $i$  divided out of the shared field in log space;
4. sum times the lattice spacing, and normalize the vector.

The cost is  $O(nL)$ : the field is built once, not once per contestant. Log-space arithmetic keeps hopeless contestants finite (their  $e^{L - \log S_i}$  underflows gracefully rather than dividing zero by zero), and the exponent is clipped at  $-745$ .

The lattice itself is placed on the winner’s bulk, not the ability span. A hopeless contestant wins only by running a winner-class time, so only the region where the winning time lives matters. Bisect the conservative winner c.d.f. envelope  $G(x) = 1 - \prod_j S_j(x)$  to  $[G^{-1}(\delta), G^{-1}(1 - \delta)]$  and pad two standard deviations, which makes the truncated tail mass negligible against the accuracy floors reported below. Measured: 33 points on this window reach accuracy that a 500-point ability-span lattice cannot, because the span window’s own truncation floors near  $6 \times 10^{-11}$ ; on fields with many hopeless contestants the bulk window is one hundred times more accurate at equal budget.

Two byproducts are cheap once the field is built. Any specified removal counterfactual, the probability  $j$  wins after deleting  $i$ , costs one more division out of the field, and any specified photo-finish tie density is one more lattice sum; §5 identifies the latter with the off-diagonal Jacobian. The full  $n \times n$  matrices of such queries are  $O(n^2)$  outputs and are priced per query, not free.

### 4 The covariance grammar

Three structures make contestants conditionally independent given a few shared variables, in increasing order of hierarchy: a global factor, factors per cluster, and a tree of uniform shared effects. Each keeps the shared field and the  $O(nLQ)$  bill; each is exact, not fitted. The approximate reduction of a dense covariance to this family is a separate matter and has its own section (§6).

## 4.1 Factor

Let  $X = \mu + Vf + \text{diag}(\sqrt{D})\epsilon$  with  $f \sim N(0, I_k)$  independent of  $\epsilon$ , so  $\Sigma = VV' + \text{diag}(D)$ . Conditional on  $f$  the performances are independent with means  $\mu + Vf$ , and

$$p = \sum_q w_q p^{\text{ind}}(\mu + Vf_q), \quad (3)$$

a  $Q$ -node quadrature over runs of the independent algorithm, cost  $O(nLQ)$ . In English: average the independent race over the shared luck. Gauss–Hermite nodes serve  $k \leq 2$ ; scrambled Sobol serves  $k \geq 3$ . Conditioning on the factor is not new: Butler and Moffitt (1982) built exactly this quadrature for the one-factor random-effects probit, and the `lprR`/`slprR` interfaces of `mvtnorm` integrate over the factor dimensions of  $BB' + \text{diag}(D)$  by Monte Carlo. What those constructions price is one alternative per integral. The field is the new part: one pass shares the survival product across all  $n$  alternatives, and §7 measures the difference against the per-alternative version of this same quadrature.

One default matters. When  $D_i \ll \|V_i\|^2$  the conditional race is nearly deterministic and the integrand over  $f$  is nearly a step function; a fixed 15-node rule then silently loses up to five percent of total variation, and the loss is invisible to lattice refinement because it is points-invariant. Randomized invariant testing against a QMC referee found this. The default order now scales with the sharpness  $\max_i \|V_i\|/\sqrt{D_i}$ , capped by rank.

The same pass returns the own-parameter slopes  $\partial p_i/\partial \mu_i$  at no extra cost (negative in this min-wins convention: raising a time mean lowers the win probability), which §5 uses as the Newton preconditioner.

## 4.2 Blocks and nested

Assign contestant  $i$  to cluster  $c(i)$  with loading  $v_i \in \mathbb{R}^r$  and let

$$X_i = \mu_i + v_i' a_{c(i)} + \sqrt{D_i} \epsilon_i, \quad a_c \text{ i.i.d. } N(0, I_r), \quad (4)$$

a block-diagonal rank- $r$ -plus-diagonal covariance: teams, conferences, sibling products. Two independences do the work. Conditional on its own cluster's effect a contestant is independent of its teammates; and distinct clusters are independent outright. So the joint survival factorizes across clusters,

$$\Pr\{X_j > x \text{ for every } j\} = \prod_c G_c(x), \quad G_c(x) = \mathbb{E}_a \prod_{j \in c} S_j(x | a) = \sum_q w_q e^{S_c(x, a_q)}, \quad (5)$$

with  $S_c(x, a) = \sum_{j \in c} \log S_j(x | a)$  the cluster's conditional log-survival. In English:  $G_c(x)$  is the probability that every member of cluster  $c$  survives past  $x$ , with the cluster's shared luck integrated out.

Contestant  $i$  then wins against a cavity field:

$$p_i = \int h_i(x) \prod_{c \neq c(i)} G_c(x) dx, \quad h_i(x) = \sum_q w_q f_i(x | a_q) e^{S_{c(i)}(x, a_q) - \log S_i(x | a_q)}. \quad (6)$$

The term  $h_i$  couples  $i$ 's win density to its own teammates within the shared draw of  $a_{c(i)}$ ; the cavity product supplies the rest of the field. The cost is  $O(nLQ_a)$  whatever the number of clusters, because the per-cluster factors are accumulated once. Rank two is special: the conditional integrand is smooth and a tensor Gauss–Hermite rule beats scrambled Sobol at equal nodes by a wide measured margin.

*Nested.* Add one global factor with couplings  $g_i$  and a dial  $\gamma \in [0, 1]$ : conditional on the global draw  $f_q$  the model is a block race at shifted abilities, so  $p = \sum_q w_q p_{\text{block}}(\mu + \gamma g f_q)$ , a finite mixture of block races. At  $\gamma = 0$  the clusters are isolated, at  $\gamma = 1$  fully coupled, and the covariance contribution scales as  $\gamma^2 g g'$ .

### 4.3 Trees

Let a rooted tree have the leaf clusters at its bottom and internal nodes  $t$  above, and give node  $t$  a scalar effect  $b_t \sim N(0, 1)$  applied with uniform strength  $\lambda_t$  to every leaf beneath it:

$$X_i = \mu_i + v_i a_{c(i)} + \sum_{t \in \text{anc}(c(i))} \lambda_t b_t + \sqrt{D_i} \epsilon_i. \quad (7)$$

The implied covariance is hierarchical: two contestants share the variance of exactly their common ancestors. The root's effect is common to every contestant, hence choice-irrelevant, and is fixed to zero. The uniform strength is the crucial restriction, because a shared effect that hits every subtree member equally moves the whole subtree's field by a translation of its argument. Messages therefore remain functions of one variable. Per-leaf loadings on internal effects would force two-dimensional message tables.

The algorithm is two passes over the same lattice.

*Upward.* Each leaf cluster carries its block factor  $G_c(x) = \sum_q w_q e^{S_c(x, a_q)}$  from (5). Each internal node, visited deepest first, combines its children under its own effect:

$$G_t(x) = \sum_q w_q \prod_{c \in \text{children}(t)} G_c(x + \lambda_t a_q). \quad (8)$$

In English:  $G_t(x)$  is the probability that every leaf below  $t$  survives past  $x$ , with  $t$ 's effect integrated by quadrature and each child's message shifted because the effect moves that child's whole subtree at once. Shifted evaluations are linear interpolations on the lattice.

*Downward.* The root's cavity is one. Each node, visited shallowest first, receives the field of everything outside its subtree:

$$R_t(x) = \left[ \sum_q w_q R_{\text{parent}(t)}(x - \lambda_{\text{parent}(t)} a_q) \right] \prod_{s \in \text{siblings}(t)} G_s(x). \quad (9)$$

The sibling messages sit OUTSIDE the parent-factor quadrature, and correctly so, though it is not obvious: the parent effect shifts the candidate and every sibling subtree by the same amount, so it cancels from every candidate-versus-sibling comparison before those comparisons are evaluated; only the cavity from outside the parent must be shifted and averaged. Contestant  $i$  then wins against its own leaf cavity,  $p_i = \int h_i(x) R_{c(i)}(x) dx$  with  $h_i$  as in (6). The cost is  $O(nLQ)$  plus  $O(LQ)$  per tree node.

A depth-one tree with zero strengths reproduces the block race to machine precision, which is tested. Against a  $2^{22}$ -path common-random-numbers referee the recursion sits at the simulation noise floor on every resolvable entry at depths one through four (root-mean-square  $z$  between 0.75 and 1.1). Entries below the referee's resolution agree with minimax tilting (Botev, 2017) to 0.3 percent relative at probabilities down to  $7 \times 10^{-8}$ , which is within the

tilting estimate’s own error. Given adequate factor quadrature, tail accuracy is carried by the lattice, and the grammar kernels price tails that simulation does not resolve.

**Remark 1** (A connection to financial practice). *Long-only portfolio weights could be considered as race probabilities, and there is a tree-race covariance under which hierarchical risk parity’s bisection geometry (López de Prado, 2016) is exactly a race: give internal node  $t$  strength  $\lambda_t^2 = \rho_t - \rho_{\text{parent}(t)}$  with  $\rho_t = \max(1 - 2h_t^2, 0)$  for merge height  $h_t$ , and leaf  $i$  idiosyncratic variance  $1 - \rho_{\text{first}(i)}$ ; the implied correlation equals the floored cophenetic matrix, with the root effect entering at its cophenetic value (when the dendrogram’s root correlation is positive and the race construction instead fixes the root effect to zero — as the shipped `from_linkage` does, the root being choice-irrelevant — the two models are equivalent on contrast space, pairwise difference variances unchanged, but the zero-root model’s RAW correlations are  $(\rho_{ij} - \rho_0)/(1 - \rho_0)$ , not  $\rho_{ij}$ ). The floor is forced, because uniform shared effects cannot represent negative dependence, and omitting it silently inflates other implied correlations above one. The identity turns dendrogram-consistent concentration limits into contest inversions, but portfolio applications are not this paper’s subject.*

## 5 Jacobians, tie densities, inversion

**Proposition 1** (Off-diagonals are photo-finish densities). *For the race (2) and  $j \neq i$ ,*

$$\frac{\partial p_i}{\partial \mu_j} = \int f_i(x) f_j(x) \prod_{k \neq i, j} S_k(x) dx \geq 0, \quad (10)$$

*and each row of the Jacobian sums to zero.*

The zero row sum says a common shift moves nothing. The off-diagonal is the density of a dead heat between  $i$  and  $j$  ahead of the field. The Jacobian is a negative weighted graph Laplacian and is applied as an operator: individual entries and Jacobian–vector products are byproducts of the same field pass (per quadrature node a Gram structure over the lattice), while materializing all  $n^2$  entries is a choice with  $\Omega(n^2)$  cost that nothing downstream requires. Three derivative objects should be distinguished: the continuum identity (10); its quadrature evaluation; and the exact derivative of the normalized, adaptive-lattice numerical map, which differs from the first two through the normalization quotient and grid motion. The implementation exposes the frozen-grid form, which matches central finite differences of the returned map WITH THE GRID HELD FIXED at the base point and the normalization quotient included, at every lattice resolution tested; a finite difference that lets the adaptive grid move with the perturbation measures a different (also legitimate) derivative and differs from the frozen-grid form by the grid-motion term. The continuum form, whose symmetry serves preconditioning, is exposed alongside. The stable pair factorization is  $[f_i e^{L - \log S_i}] \times [f_j / S_j]$ : the first factor is bounded, and the hazard  $f_j / S_j$  grows linearly in the normal base and exponentially in the Gumbel base, either of which the exponent clip absorbs. The symmetric square-root split overflows where survivals vanish.

The block Jacobian is exact, with the within-cluster term computed under the cavity; the nested Jacobian is the mixture of block Jacobians; the tree Jacobian is exact within a cluster and Gram-approximate across clusters, which suffices for Newton because residuals always use the exact forward map. Feasibility-critical callers fall back to finite differences.

The inversion rests on a convex program.



**Theorem 1** (The inverse exists and is unique on contrasts). *Fix a covariance of full support (any member of the grammar with every  $D_i > 0$ ) and let  $W(\mu) = \mathbb{E} \min_i (\mu_i + \epsilon_i)$ ,  $\epsilon \sim N(0, \Sigma)$ . Then  $W$  is concave and differentiable with  $\nabla W(\mu) = p(\mu)$ , and its Hessian is  $-L(\mu)$ , the negative of the graph Laplacian whose edge weights are the tie densities (10). Every tie density is strictly positive, so  $W$  is strictly concave on the contrast space  $\mathbf{1}^\perp$ . For any target with  $p_i^* > 0$  and  $\sum_i p_i^* = 1$ , the program*

$$\max_{\mu} W(\mu) - \langle p^*, \mu \rangle$$

*is constant along  $\mathbf{1}$ , strictly concave and coercive on  $\mathbf{1}^\perp$ , and its unique mean-zero maximiser is the inverse:  $p(\mu^*) = p^*$ . If some  $p_i^* = 0$  the supremum is approached only as the contrast  $\mu_i - \min_{j: p_j^* > 0} \mu_j \rightarrow +\infty$  (the raw coordinate statement depends on the gauge); no finite inverse exists, and a floored target recovers a lower bound on that contrast.*

*Proof.* A minimum of affine functions of  $\mu$  is concave and expectation preserves this. Ties have measure zero under full support, so the minimiser is almost surely unique and Danskin's theorem gives  $\partial W / \partial \mu_i = \Pr\{i \text{ wins}\}$ .

The Hessian requires no independence assumption, and we derive it in three steps. Off-diagonal ( $j \neq i$ ): raising  $\mu_j$  moves only  $X_j$ , and the derivative of  $p_i = \Pr\{X_i \leq X_k \ \forall k\}$  is the probability flux through the boundary  $\{X_j = X_i\}$  of the winning region,

$$\frac{\partial p_i}{\partial \mu_j} = \int \varphi_{ij}(x, x) \Pr\{X_k > x, \ k \neq i, j \mid X_i = X_j = x\} dx \geq 0,$$

with  $\varphi_{ij}$  the joint density of  $(X_i, X_j)$  under the full correlated law: the photo-finish density of  $i$  and  $j$  jointly ahead of the field. Diagonal: translation invariance  $W(\mu + c\mathbf{1}) = W(\mu) + c$  forces the gradient to sum to one, hence every Hessian row to sum to zero, so  $\partial p_i / \partial \mu_i = -\sum_{j \neq i} \partial p_i / \partial \mu_j$  with no further computation. Reduction: when the law is conditionally independent given shared variables—every grammar member—conditioning factorizes the flux integral into (10)'s product form node by node. The Hessian is therefore  $-L(\mu)$  with  $L$  a weighted graph Laplacian; under full support ( $D_i > 0$ , so every pairwise difference has positive variance) each edge weight is strictly positive, the tie graph is complete, and  $L \succ 0$  on  $\mathbf{1}^\perp$ :  $W$  is strictly concave there.

Coercivity follows from the elementary bound  $W(\mu) \leq \min_i \mu_i$ : for each  $i$ ,  $\min_k (\mu_k + \epsilon_k) \leq \mu_i + \epsilon_i$ , and  $\mathbb{E} \epsilon_i = 0$ . For a nonconstant unit contrast  $d$  and interior target,  $\langle p^*, d \rangle > \min_i d_i$ , because  $p^*$  puts positive mass on a coordinate where  $d$  exceeds its minimum; hence

$$W(td) - t\langle p^*, d \rangle \leq t(\min_i d_i - \langle p^*, d \rangle) \longrightarrow -\infty$$

linearly in  $t$ . The decay is uniform, not merely raywise:  $d \mapsto \langle p^*, d \rangle - \min_i d_i$  is continuous and strictly positive on the compact unit sphere of  $\mathbf{1}^\perp$ , so it attains a positive minimum  $\kappa$ , and the objective is bounded above by  $W(0) - \kappa\|\mu\|$  along every direction — which is coercivity for arbitrary diverging sequences. Constancy along  $\mathbf{1}$  is translation invariance again, exactly when the target sums to one. A strictly concave coercive function on  $\mathbf{1}^\perp$  attains a unique maximum, where the gradient vanishes:  $p(\mu^*) = p^*$ .  $\square$

The program is the Legendre dual of the social surplus function of McFadden (1981), and share inversion as a convex problem is established territory: Berry (1994) inverts market

shares in logit-family demand, Galichon and Salanié (2022) identify the general random-utility inversion with optimal transport, and Chiong et al. (2016) compute it by linear programming in the discrete case. Closest to an algorithm is Li (2018), who for arbitrary non-atomic bases writes the inversion as an unconstrained convex minimization whose objective gradient is the share map, with globally convergent superlinear methods — the formulation and the optimizer, without a fast evaluator of the shares and their Jacobian at large  $n$ . The shared field is that evaluator, so the two are complements. What the field representation adds is not the convex program but its derivatives: the Hessian is the tie-density Laplacian of the same field pass, so gauge-fixed Newton runs matrix-free at  $O(nLQ)$  per iteration.

Identification lives on the same contrast space. Choice probabilities are invariant to a common utility shock, so the model depends on  $\Sigma$  only through  $P\Sigma P$  with  $P = I - \mathbf{1}\mathbf{1}^\top/n$ : a factor proportional to  $\mathbf{1}$ , or a tree-root effect applied to every leaf, is choice-irrelevant. The right fitting objective for a dense covariance is therefore the projected residual,

$$\min_{V, D \geq 0} \|P(\Sigma - VV' - \text{diag } D)P\|_F^2,$$

not a fit of  $VV' + \text{diag}(D)$  to  $P\Sigma P$  (the projection of a rank-plus-diagonal matrix is no longer rank-plus-diagonal, so the second problem manufactures error the first does not have). The reference implementation minimizes the stated objective (`factor_model_projected`: alternate a top- $k$  PSD step for  $V$  in the contrast basis with the exact nonnegative diagonal solve, whose normal equations are  $(P \circ P)d = \text{diag}(P(\Sigma - VV')P)$ ), and §6’s pipeline consumes only projected residuals downstream, so an input already of the form  $VV' + \text{diag}(D)$  is returned undistorted. The introduction’s identified-class sentence is to be read in this sense: structured covariance is the estimable and computable restriction on  $\mathbf{1}^\perp$ , not a claim that structure exhausts identification.

The shipped iteration is stated so it can be checked against the code (`abilities_from_race`).

**Inversion** (all grammars; min-wins). Given a target  $p^\star$  with positive entries summing to one, set  $t = \log p^\star$  and initialize  $\mu^0 = -(t - \bar{t})/2$ . Repeat until  $\max_i |r_i| < 10^{-8}$ :

1. one field pass:  $\hat{p} = p(\mu)$ , and own-slopes  $s_i = \partial p_i / \partial \mu_i$  (factor grammar: the same pass; block/nested/tree: the slopes of the independent race at matched total variances, one extra  $O(nL)$  pass);
2. log residual  $r_i = \log \hat{p}_i - t_i$ ; log-domain slope  $d_i = \min(s_i / \hat{p}_i, -10^{-6})$ ;
3. damped diagonal Newton step  $\mu_i \leftarrow \mu_i - \text{clip}(\alpha r_i / d_i, \pm 2)$ , then recenter  $\mu \leftarrow \mu - \bar{\mu}$ .

Damping:  $\alpha = 1$  (0.7 at  $n = 2$ , where the undamped update on the  $K_2$  photo-finish graph is a two-cycle); for the structure grammars  $\alpha = 0.7$ , halved whenever the residual fails to shrink.

This is a diagonally preconditioned Newton iteration on log probabilities; Theorem 1 makes the target a well-posed problem, and the clip and recentering confine the iterate to the contrast space. To prevent overreading: the theorem establishes well-posedness, not convergence of this particular clipped iteration, which is an empirically safeguarded solver — the stress suite converges everywhere tested, and a safeguarded convex or Newton–Krylov fallback is available in principle should a case ever demand it. Round trips through the shipped front door at  $n = 400$  (the committed `bench.py invert`; twenty clusters, tree of depth three): independent

0.04 s, tree 0.37 s, blocks 0.67 s, factor rank-2 1.7 s, nested 3.7 s, each to a maximum log-probability residual below  $10^{-8}$ . Targets below the forward map’s resolution are floored, with the one-sided bound direction the theorem states. Ignoring known structure at inversion time is not a small error: inverting block-generated probabilities under an assumed independent model mislocates abilities by up to three tenths of the field’s spread (the pinned configuration:  $n = 200$ , thirty clusters, within-cluster correlation 0.65; maximum mislocation 2.49 against a spread of 8.24, median 0.34).

## 6 Dense covariance

The fitting procedure, stated so it can be reproduced. The fit runs in COVARIANCE coordinates, deliberately: an earlier draft standardized to a correlation matrix, fit there, and restored scales afterwards, and a referee supplied the two-line refutation. With  $\Sigma = SCS$ ,  $S = \text{diag}(s)$ , the choice-relevant residual after restoration is  $PS(C - M)SP$ , not  $P(C - M)P$ , and  $P$  does not commute with an unequal  $S$ : at  $n = 2$  with equicorrelated  $C$  ( $\rho = 0.9$ ) and  $s = (1, 2)$ , the correlation-space quotient objective declares the fit  $M = (1 - \rho)I$  exact while the restored model gets the only identified quantity, the difference variance, wrong by nearly a factor of three (0.5 against 1.4). Equivalently, the choice-irrelevant direction in correlation coordinates is  $S^{-1}\mathbf{1}$ , not  $\mathbf{1}$ . A single global rescaling for conditioning is harmless; per-coordinate standardization is not, and the shipped test suite pins the analytic two-alternative example and an exact-grammar round trip with variances spanning two decades. So: fit  $\Sigma$  itself. (i) Global factors: the quotient fit of §5 (`factor_model_projected`),  $k = 3$  throughout, returning  $(V, D_0)$  by alternation that decreases the identified objective monotonically to a coordinatewise-stationary point (the objective is nonconvex in the pair, so no global certificate is claimed; multistart dispersion is a cheap diagnostic and the residual report exposes the value reached). (ii) The working residual is then projected once,  $R = P(C - VV' - \text{diag } D_0)P$ : the raw residual still contains the choice-irrelevant common component the quotient fit rightly ignored, and letting later stages chase it manufactures projected error (measurably: an input already of the form  $VV' + \text{diag}(D)$  came back with total variation  $1.7 \times 10^{-2}$  under the raw-residual pipeline, and returns at the  $2 \times 10^{-4}$  node-noise floor under the projected one). (iii) Blocks: average-linkage clustering on  $\sqrt{(1 - C)/2}$  cut to 20 clusters; for each cluster, the leading eigenvector of the within-cluster block of  $R$  with its diagonal zeroed, one rank-one loading per cluster (clusters are disjoint). (iv) Residual promotion: the top  $m = 5$  eigencomponents of what remains of  $R$ , diagonal zeroed, appended as further factor columns; negative eigenvalues are not promoted; numerically dead columns are dropped. (v) The closing diagonal solves the identified problem’s own normal equations,  $(P \circ P)d = \text{diag}(P(C - V_{\text{all}}V'_{\text{all}})P)$ , floored at  $10^{-3}$  (the Gram is exactly  $(1 - \frac{2}{n})I + \frac{1}{n^2}\mathbf{1}\mathbf{1}^\top$ , so both the solve and its nonnegative version are closed-form). (vi) A second candidate — a pure eigenfit at the same total rank, closed by the same diagonal solve — is computed, and whichever candidate leaves the smaller choice-relevant residual is kept, the pipeline winning ties: the greedy factor-plus-blocks allocation is the wrong shape for globally smooth covariance, and the kernel battery below measures the repair. The constants  $k$ , the cluster count and  $m$  are fixed, not tuned per ensemble; a spectral rule for  $m$  was tested and rejected (it saturates its cap for marginal gains and does not repair the failure case below). What results is a factor-plus-blocks race priced in one deterministic call, and the pipeline is the shipped one-call (`fit_covariance`; `race_probabilities` accepts `cov=` directly), not a research script. The shipped call also reports its own quality and warns on two distinct failure modes:

a large choice-relevant residual (the fit could not hold the matrix), and a well-fitted matrix whose pricing needs a high-rank, sharp factor integral (the quadrature, not the fit, is then the binding constraint; both regimes appear in the kernel battery below). At  $n = 2000$  the one-call fit-and-price takes four seconds (0.6 of them the fit) against thirty-six for a  $10^6$ -draw simulation, agrees with it at its noise floor, and prices the 82 percent of tail alternatives for which the simulation records zero wins.

An earlier draft compared a raw-space fit against a fit of  $VV' + \text{diag}(D)$  to  $P\Sigma P$  and found the latter worse on several ensembles; a referee correctly identified that as a comparison between the raw objective and the *wrong* projected objective ( $P\Sigma P$  of a rank-plus-diagonal matrix is not rank-plus-diagonal). With the identified objective of §5 and projected residuals throughout, the ensemble study below ran both arms — a raw top- $k$  eigenfit pipeline and the identified pipeline — over twenty seeds per ensemble. The measured outcome: on thirteen of fifteen named ensembles the arms are statistically indistinguishable, none favors the raw arm, and on block equicorrelation the identified arm is eight times better at the median ( $2.5 \times 10^{-3}$  against  $2.0 \times 10^{-2}$ ) with a worst seed of  $3.2 \times 10^{-3}$  against the raw arm’s 0.162: the raw eigenfit spends its rank on a near-common component exactly as the identification argument predicts. The ensembles the earlier draft reported as worse under projection (factor with sparse links, hierarchical) are modestly *better* under the correct objective.

Accuracy depends on the generating ensemble, so it is reported per named ensemble of the `randomcov` library (pinned at commit `0d27a51`), now over *twenty seeds* per ensemble at  $n = 300$ , each seed refereed by its own  $10^6$ -path simulation (the committed `run_ensembles4.py` and `run_kernel14.py` regenerate everything). To be unambiguous about the statistic: the TV columns are FULL-vector total variation against the empirical frequency vector, zeros included, so the referee’s own counting noise and any unresolved tail mass contribute to the reported number (they bound it from above); only the per-entry column is restricted, to entries with at least 25 referee wins. TV is reported as median, ninth decile and worst over seeds; the last column is the median per-entry absolute error on those resolved entries. Representative rows, identified pipeline:

| ensemble                   | med TV               | q90 TV               | worst TV             | med $ p - \hat{p} $  |
|----------------------------|----------------------|----------------------|----------------------|----------------------|
| block equicorrelation      | $2.5 \times 10^{-3}$ | $3.0 \times 10^{-3}$ | $3.2 \times 10^{-3}$ | $1.4 \times 10^{-5}$ |
| factor + sparse links      | $2.9 \times 10^{-3}$ | $6.7 \times 10^{-3}$ | $1.6 \times 10^{-2}$ | $1.9 \times 10^{-5}$ |
| sparse-precision graphical | $4.5 \times 10^{-3}$ | $5.6 \times 10^{-3}$ | $6.0 \times 10^{-3}$ | $1.5 \times 10^{-5}$ |
| Wishart                    | $1.2 \times 10^{-2}$ | $1.4 \times 10^{-2}$ | $1.5 \times 10^{-2}$ | $2.6 \times 10^{-5}$ |
| AR(1)                      | $4.6 \times 10^{-2}$ | $8.4 \times 10^{-2}$ | $1.5 \times 10^{-1}$ | $7.0 \times 10^{-5}$ |
| residuals (dense-strong)   | $6.8 \times 10^{-2}$ | $8.6 \times 10^{-2}$ | $9.2 \times 10^{-2}$ | $2.5 \times 10^{-4}$ |

The multi-seed medians reproduce the single-draw table of the earlier draft to within seed noise on every surviving row, so the original numbers were representative, not fortunate. Eight further named ensembles (LKJ, onion, vine, elliptope walk, animals, Marchenko–Pastur and spiked spectra, hierarchical) fall between the third and fifth rows; Archakov–Hansen sits with AR(1) at  $4.9 \times 10^{-2}$ .

The kernel family is the failure case, and the stratified battery (kernel type  $\times$  length scale  $\times$  budget, twenty seeds per cell, unit-square inputs) separates two distinct mechanisms that a single “kernel” row had conflated:

| kernel                | length scale | med TV, $m=5$        | $m=12$               | $m=5, 2^{14}$ nodes  |
|-----------------------|--------------|----------------------|----------------------|----------------------|
| RBF                   | 0.08         | $3.4 \times 10^{-2}$ | $3.1 \times 10^{-2}$ | $3.3 \times 10^{-2}$ |
| RBF                   | 0.2          | $2.6 \times 10^{-2}$ | $2.6 \times 10^{-2}$ | $1.2 \times 10^{-2}$ |
| RBF                   | 0.4          | $1.6 \times 10^{-2}$ | $1.7 \times 10^{-2}$ | $8.2 \times 10^{-3}$ |
| Matérn- $\frac{3}{2}$ | 0.08         | $2.9 \times 10^{-2}$ | $2.5 \times 10^{-2}$ | $2.4 \times 10^{-2}$ |
| Matérn- $\frac{3}{2}$ | 0.2          | $2.5 \times 10^{-2}$ | $2.3 \times 10^{-2}$ | $1.6 \times 10^{-2}$ |
| Matérn- $\frac{3}{2}$ | 0.4          | $2.1 \times 10^{-2}$ | $1.7 \times 10^{-2}$ | $1.2 \times 10^{-2}$ |

At short length scale the bottleneck is *representation*: correlation is locality, the rank needed grows with the number of correlation lengths in the domain, and accordingly extra rank helps ( $3.4 \rightarrow 3.1 \times 10^{-2}$ ) while extra quadrature does not. At long length scale the fit is not the problem at all — a rank-27 eigenfit holds the RBF draw at length scale 0.4 to a projected residual of  $10^{-3}$  — but pricing a well-fitted smooth kernel means a twenty-seven-dimensional *sharp* factor integral (the closing diagonal is nearly zero, so conditional races are nearly deterministic), and there the node budget is the binding constraint: quadrupling rank does nothing while  $2^{14}$  nodes halve the error, and  $2^{16}$  nodes reach  $6 \times 10^{-3}$  on the diagnostic draw. Getting even this far required the pipeline’s stage (vi): the greedy factor-plus-blocks allocation left a projected residual of 0.14 on the same draw that the matched-rank eigenfit holds to  $10^{-3}$ , and the earlier draft’s kernel row (up to  $7 \times 10^{-2}$ ) was that misallocation, not a limit of the grammar. What remains true is the dispatch conclusion: genuinely local covariance wants sequential conditioning with cheap sparse-precision draws — Matérn is exactly the family with the SPDE route (Lindgren et al., 2011) — near-singular smooth covariance wants either a larger node budget or plain simulation, and a production front door should dispatch on the structure it detects; the shipped call’s warnings distinguish the two regimes.

## 7 Benchmarks

*Against the field.* All- $n$  choice probabilities under the rank-2 factor covariance family of Section 2, against a  $2^{20}$ -point QMC reference. Tail is the maximum absolute log-error over reference probabilities in  $[10^{-4}, 10^{-3}]$ ; the  $n = 10$  draw has none. Frequency simulation is run at wall clock matched to the lattice and at ten times that budget; “zeros” counts tail alternatives that received no draws.

| method                     | $n = 10$ |                      | $n = 30$ |                      | tail          |
|----------------------------|----------|----------------------|----------|----------------------|---------------|
|                            | time     | TV                   | time     | TV                   |               |
| lattice race               | 2.5 ms   | $1.8 \times 10^{-4}$ | 4.7 ms   | $8.3 \times 10^{-4}$ | 0.07          |
| Genz, full vector          | 46 ms    | $1.8 \times 10^{-4}$ | 1.40 s   | $8.3 \times 10^{-4}$ | 0.07          |
| frequency, matched time    | —        | $3.8 \times 10^{-3}$ | —        | $1.0 \times 10^{-2}$ | 0.45, 2 zeros |
| frequency, $10\times$ time | —        | $9.2 \times 10^{-4}$ | —        | $2.0 \times 10^{-3}$ | 0.21          |
| Mendell–Elston             | 2.0 ms   | $1.8 \times 10^{-3}$ | 18 ms    | $3.2 \times 10^{-2}$ | 1.30          |
| Clark-type                 | 1.4 ms   | $1.1 \times 10^{-2}$ | 17 ms    | $1.6 \times 10^{-2}$ | 1.33          |

Genz and the lattice agree to the reference’s noise and to each other. Of the  $300\times$  cost ratio at  $n = 30$ , the one-at-a-time interface supplies an unavoidable  $n$ -fold component; the remainder reflects the differing integral representations, tolerances and implementations, and we do not apportion it further. At the paper’s headline scale the  $n$ -fold component alone is decisive: one probability at  $n = 200$  costs 0.43 s at `maxpts` =  $2.5 \times 10^5$  (agreement with the

lattice to four to five digits; six digits needs multi-second budgets — the committed `bench.py` (`genz200`), so the full vector costs 86 s against 26 ms for the lattice, a factor of  $3 \times 10^3$  before any accuracy matching. The sequential-conditioning family is the only entry whose error grows silently: its tail entries are off by  $e^{1.3} \approx 3.7$  with nothing in the output to say so. The race’s tail figure is the reference’s own noise; lattice self-convergence places its error near  $10^{-8}$ .

*Referee agreement.* A replay harness maps each benchmark probability to its difference or-thant and prices it through the two field-standard R implementations. `mvtnorm`’s Genz–Bretz agrees with the lattice to within its own reported error bound in every family:  $3.8 \times 10^{-7}$  against a bound of  $8.1 \times 10^{-7}$  at  $n = 10$ , and  $2.7 \times 10^{-8}$  against  $5.8 \times 10^{-8}$  on a case whose smallest probability is  $3 \times 10^{-10}$ . Botev’s minimax tilting agrees to its own Monte Carlo noise, which in relative terms is  $2.2 \times 10^{-4}$  at  $p \approx 3 \times 10^{-10}$ : the tail claim rests on the named accuracy reference, not only on self-convergence. On the independent family an adaptive-quadrature referee reaches machine precision and the lattice matches it to  $1.9 \times 10^{-15}$ . An invariance battery (two-runner closed form, translation, permutation, symmetry) holds to  $10^{-16}$  throughout, except the two-runner comparison at  $7 \times 10^{-9}$ , which is the lattice discretization itself.

*Stress boundaries.* An adversarial battery pushes each failure axis until something gives. Depth: against pure-relative adaptive quadrature, relative error is  $4 \times 10^{-13}$  at  $p \approx 10^{-30}$  and  $2 \times 10^{-8}$  at  $p \approx 10^{-50}$ , with the first genuine breakdown near  $p \approx 10^{-119}$ , a laggard twenty-five standard deviations behind. Inversion: the round trip  $p \rightarrow \mu \rightarrow p$  holds  $10^{-9}$  in log even when the target contains a  $10^{-10}$  entry or ties at the  $10^{-9}$  level. The Gumbel base reproduces softmax to  $10^{-15}$  at  $n = 1000$ ; exact duplicates split exactly, and an  $\epsilon$  perturbation moves the split linearly. The one operative boundary is factor strength. When loadings make implied correlations exceed roughly 0.9, the argmax integrand loses smoothness in factor space and Gauss–Hermite converges slowly in the node count: at loading scale 4 the 25-node rule still sits at  $8 \times 10^{-3}$  total variation. Scrambled-Sobol factor nodes restore reference-level accuracy ( $4 \times 10^{-4}$ ) at the same call: heavy correlation wants quasi-Monte-Carlo nodes, not deeper polynomial rules. The default now escalates the family automatically past sharpness 3, in both the Python and R implementations (scrambled Sobol and Halton respectively), which agree to  $5 \times 10^{-4}$  on the worst case; the figures above are the pre-escalation diagnosis.

*Against GHK.* All- $n$  choice probabilities under a rank-2 factor covariance, against a  $2^{20}$ -point QMC reference. TV is a bulk metric and blind to the tails, so the same tail column as above is reported: the maximum absolute log-error over reference probabilities in  $[10^{-4}, 10^{-3}]$ , which for probabilities this small is the log-odds error, and which the reference’s own counting noise floors near 0.1.

| $n$ | lattice race |                      |      | GHK, $R = 10^3$ |                      |      | GHK, $R = 10^4$ |                      |      |
|-----|--------------|----------------------|------|-----------------|----------------------|------|-----------------|----------------------|------|
|     | time         | TV                   | tail | time            | TV                   | tail | time            | TV                   | tail |
| 10  | 2.3 ms       | $1.8 \times 10^{-4}$ | —    | 1.0 ms          | $8.4 \times 10^{-3}$ | —    | 8 ms            | $2.1 \times 10^{-3}$ | —    |
| 50  | 7.0 ms       | $1.7 \times 10^{-3}$ | 0.11 | 21 ms           | $4.2 \times 10^{-2}$ | 1.10 | 214 ms          | $1.2 \times 10^{-2}$ | 0.29 |
| 200 | 25 ms        | $2.4 \times 10^{-3}$ | 0.15 | 436 ms          | $3.2 \times 10^{-2}$ | 1.50 | 4.8 s           | $1.2 \times 10^{-2}$ | 0.35 |

The  $n = 10$  draw has no reference probability in the tail band. The lattice’s tail figures equal the reference’s own noise; GHK’s are two to ten times larger in log terms even at ten times the draws, which is the tail blindness the TV column cannot show.

The lattice bounds are the reference’s own noise; self-convergence places the lattice error near  $10^{-8}$ . GHK error contracts as  $O(R^{-1/2})$  and its measured complete-vector cost grows superquadratically, with exponent 2.15 to 2.8 across the measured range. The lattice is deterministic, supplies smooth matrix-free derivatives free of simulation noise (simulated likelihoods can be differentiated, but inherit the noise), and prices the tails. The claim, stated so it cannot be read past its evidence: for complete-vector pricing and inversion under the factor, block, nested and tree covariance grammars, the shared-field construction removes the per-alternative repetition intrinsic to standard GHK implementations and dominates the per-alternative GHK implementation tested here in both accuracy and cost. GHK implementations with antithetics, quasi-random sequences or reordering heuristics would move the constants, not the  $n$ -fold shape. Estimation pipelines built around GHK are a larger object than this loop, and §7’s estimation paragraphs report where simulation remains competitive at face value. GHK also retains general rectangle probabilities and locality-structured covariance, and Botev (2017) remains the reference for single extreme tail probabilities.

*The structure-aware per-alternative comparator.* The fairest rival is not GHK but this paper’s own conditioning run the literature’s way: for each alternative separately, integrate the conditional independent-race integrand over the same Gauss–Hermite factor nodes (the construction of Butler and Moffitt (1982); `mvtnorm`’s `lpRR`/`slpRR` are its Monte Carlo form). With the same nodes and the same lattice the two computations are algebraically identical, and measure so: at  $n = 200$ , rank 2, the per-alternative version agrees with the shared field to total variation  $3 \times 10^{-15}$  and costs 601 times as much (22 s against 36 ms; the committed `bench.py bm`). The comparison isolates the paper’s contribution exactly: not the conditioning, which is forty years old, but the field shared across alternatives.

*Share inversion under sampling noise.* The first estimation exercise is deliberately the saturated case. Simulate  $T = 50,000$  choices from one fixed menu of  $n = 30$  alternatives under a known rank-2 factor covariance and estimate the mean-zero utilities by maximum likelihood. With the menu fixed the model is saturated over the simplex interior, so the exact-gradient maximizer is the inversion of empirical shares,  $\hat{\mu} = p^{-1}(c/T)$ , and the exercise measures how faithfully each method performs that inversion under multinomial noise. Eight replications:

| estimator          | RMSE over identified alternatives | median fit time |
|--------------------|-----------------------------------|-----------------|
| exact gradient MLE | 0.0187                            | 4.9 s           |
| MSL, $R = 100$     | 0.0305                            | 4.4 s           |
| MSL, $R = 1000$    | 0.0198                            | 40.7 s          |

At equal fitting time the simulated inversion pays roughly a 60 percent error premium in this diagnostic (8 replications; the paired difference is stable in sign across all eight, but the premium’s magnitude carries replication noise we have not driven down, and we present it as a diagnostic rather than an anchored constant). The  $R = 1000$  figure is close to the exact one and the difference is within replication noise; the claim is the equal-time column, not the third row.

*Covariate estimation.* The overidentified case behaves differently, and we report it as we found it. Draw observation-specific design matrices  $X_t$  ( $n = 10$  alternatives,  $d = 3$  covariates,

$T = 800$  observations), utilities  $\mu_t = X_t\beta$ , one choice per observation, covariance known, and estimate  $\beta$  by individual-level maximum likelihood: the exact path uses lattice probabilities with the analytic score (one Jacobian row per observation, a single field pass); the simulated path uses this paper’s own Rust GHK with common random numbers and finite-difference gradients. Eight replications, means with standard errors:

| estimator          | RMSE( $\hat{\beta}$ ) | median fit time |
|--------------------|-----------------------|-----------------|
| exact gradient MLE | $0.0363 \pm 0.0066$   | 41.3 s          |
| MSL, $R = 100$     | $0.0380 \pm 0.0073$   | 5.4 s           |
| MSL, $R = 1000$    | $0.0367 \pm 0.0067$   | 39.6 s          |

All three estimators reach the statistical floor: with three parameters and eight hundred observations the simulation error averages out, and a well-implemented simulator at  $R = 100$  is the fastest of the three at this size. The exact path’s time is dominated by per-observation Python overhead (the compiled kernel accounts for under a second of the 41); the comparison at  $n = 10$  measures plumbing, not method. What the exact likelihood buys in estimation is not this row of the table: it is freedom from the choice of  $R$ , gradients without simulation noise, the cost law as  $n$  grows, and diagnostics that simulation smooths over. The last point is not hypothetical: on the classic Fishing data, the unrestricted-covariance MNP likelihood evaluated exactly is boundary-seeking (still rising at loading norms near 4000), a ridge that GHK’s simulation noise had been regularizing by accident. The behavior is reproducible from the committed estimator (`winning.mnprobit`), which detects the ridge and reports it as a `boundary_` diagnostic rather than returning the interior point simulation would have manufactured.

*The Genz–Bretz score.* Our quadrature comparator used `pmvnorm`, which exposes no score, with finite-difference gradients (31 vector evaluations per step at  $n = 30$ , roughly eight times the entire exact-gradient share-inversion fit). The `mvtnorm` manual (v1.4-2) also documents `slpmvnorm` and, for the reduced-rank covariance  $BB' + \text{diag}(D)$  of this paper’s benchmarks, `lpRR` and `slpRR`: simulated log-likelihoods and score functions that integrate over the  $K$  factor dimensions. That interface is itself the conditioning construction of §4.1 with Monte Carlo in place of quadrature and no shared field across alternatives, which makes it the natural structure-aware comparator for a fuller estimation study.

*GHK protocol.* The GHK figures throughout use this paper’s own Rust implementation: per-alternative sequential conditioning on the reference-differenced covariance with a fresh Cholesky factorization per alternative, pseudorandom draws (xoshiro family) with a per-alternative seed offset so that parameter evaluations share common random numbers, no antithetics, no quasi-random sequence, no variable-reordering heuristic, parallelized across alternatives. Cost figures are complete-vector; the large- $n$  columns of the introduction are extrapolations of the measured law and are displayed as such.

*Scale.* Ten million contestants price in 245 seconds under the block grammar (ten thousand clusters, 257-point lattice, the streaming field pass; the committed `bench.py tenmillion`), and the Rust classic-calibration kernel runs 232 times the speed of the pure-Python implementation. Throughput at this size is a systems statement; the accuracy evidence lives at the scales the referees above can reach.



*Parity*. One vector file embeds the inputs and outputs of 22 scenarios spanning every claim above. The four language implementations of §8 replay them: twenty at machine precision, the optimizer-mediated scenarios to  $10^{-6}$ .

## 8 Availability

`pip install winning` is pure Python; `winning[fast]` adds the Rust kernels as one abi3 wheel per platform. A base-R package mirrors the full interface, and a zero-dependency browser port drives a live demonstration at [winning.microprediction.org](https://winning.microprediction.org).

## References

- J. H. Albert, S. Chib. Bayesian analysis of binary and polychotomous response data. *JASA* 88:669–679, 1993.
- S. Berry. Estimating discrete-choice models of product differentiation. *RAND Journal of Economics* 25(2):242–262, 1994.
- J. S. Butler and R. Moffitt. A computationally efficient quadrature procedure for the one-factor multinomial probit model. *Econometrica* 50(3):761–764, 1982.
- Z. I. Botev. The normal law under linear restrictions: simulation and estimation via minimax tilting. *Journal of the Royal Statistical Society B* 79(1):125–148, 2017.
- A. Börsch-Supan, V. Hajivassiliou. Smooth unbiased multivariate probability simulators for maximum likelihood estimation of limited dependent variable models. *Journal of Econometrics* 58:347–368, 1993.
- M. Batram, D. Bauer. On consistency of the MACML approach to discrete choice modelling. *Journal of Choice Modelling* 30:1–16, 2019.
- D. Bauer. *Rprobit: R package for the estimation of multinomial probit models*. [github.com/dbauer72/Rprobit](https://github.com/dbauer72/Rprobit).
- C. R. Bhat. The maximum approximate composite marginal likelihood (MACML) estimation of multinomial probit-based unordered response choice models. *Transportation Research Part B* 45(7):923–939, 2011.
- L. Cappellari, S. P. Jenkins. Multivariate probit regression using simulated maximum likelihood. *Stata Journal* 3(3):278–294, 2003.
- K. Chiong, A. Galichon, M. Shum. Duality in dynamic discrete-choice models. *Quantitative Economics* 7(1):83–115, 2016.
- C. E. Clark. The greatest of a finite set of random variables. *Operations Research* 9:145–162, 1961.
- Y. Croissant. Estimation of random utility models in R: the mlogit package. *Journal of Statistical Software* 95(11):1–41, 2020.
- P. Cotton. Inferring relative ability from winning probability in multi-entrant contests. *SIAM Journal on Financial Mathematics* 12(1):295–317, 2021.
- A. Galichon, B. Salanié. Cupid’s invisible hand: social surplus and identification in matching models. *Review of Economic Studies* 89(5):2600–2629, 2022.

- R. Gates. A Mata Geweke–Hajivassiliou–Keane multivariate normal simulator. *Stata Journal* 6(2):190–213, 2006.
- A. Genz. Numerical computation of multivariate normal probabilities. *Journal of Computational and Graphical Statistics* 1:141–149, 1992.
- J. Geweke. Bayesian inference in econometric models using Monte Carlo integration. *Econometrica* 57:1317–1339, 1989.
- V. Hajivassiliou, D. McFadden, P. Ruud. Simulation of multivariate normal rectangle probabilities and their derivatives. *Journal of Econometrics* 72:85–134, 1996.
- K. Imai, D. A. van Dyk. MNP: R package for fitting the multinomial probit model. *Journal of Statistical Software* 14(3):1–32, 2005.
- M. P. Keane. A computationally practical simulation estimator for panel data. *Econometrica* 62:95–116, 1994.
- S. Lerman, C. Manski. On the use of simulated frequencies to approximate choice probabilities. In *Structural Analysis of Discrete Data*, MIT Press, 1981.
- L. Li. A general method for demand inversion. arXiv:1802.04444 (2018).
- F. Lindgren, H. Rue, J. Lindström. An explicit link between Gaussian fields and Gaussian Markov random fields. *Journal of the Royal Statistical Society B* 73(4):423–498, 2011.
- R. Loaiza-Maya, D. Nibbling. Scalable Bayesian estimation in the multinomial probit model. *Journal of Business & Economic Statistics* 40(4):1678–1690, 2022.
- M. López de Prado. Building diversified portfolios that outperform out of sample. *Journal of Portfolio Management* 42(4):59–69, 2016.
- D. McFadden. Econometric models of probabilistic choice. In C. Manski, D. McFadden, eds., *Structural Analysis of Discrete Data*, MIT Press, 1981.
- R. McCulloch, P. Rossi. An exact likelihood analysis of the multinomial probit model. *Journal of Econometrics* 64:207–240, 1994.
- N. Mendell, R. Elston. Multifactorial qualitative traits: genetic analysis and prediction of recurrence risks. *Biometrics* 30:41–57, 1974.
- A. R. Solow. A method for approximating multivariate normal orthant probabilities. *Journal of Statistical Computation and Simulation* 37:225–229, 1990.
- S. Stern. A method for smoothing simulated moments of discrete probabilities in multinomial probit models. *Econometrica* 60:943–952, 1992.
- StataCorp. *Stata Choice Models Reference Manual: cmmprobit, asmprobit*. Stata Press, College Station, 2023.
- L. L. Thurstone. A law of comparative judgment. *Psychological Review* 34:273–286, 1927.
- K. Train. *Discrete Choice Methods with Simulation*. 2nd ed., Cambridge, 2009.